

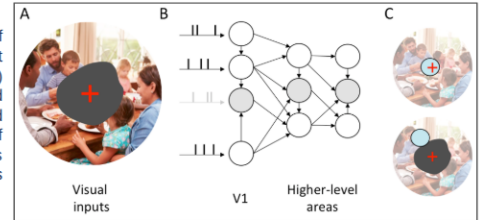
Modelling Synaptic Reorganization Following Macular Degeneration In Biologically Constrained Deep Neural Networks

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Work in brief: The aim of this project is to develop computational models that explain the evolution of the visual system following the onset of macular degeneration. To achieve this, we utilize artificial neural networks pre-trained on images or videos, which are then fine-tuned using an artificial scotoma to simulate central vision loss. After training, the network's properties are analyzed using explainable AI (xAI) techniques and compared with biological measurements.

Summary: Age-related macular degeneration (AMD) is a neurodegenerative condition marked by the gradual loss of photoreceptors in the fovea and which currently affects over 65 million of patients in Europe (Li et al., 2020). Following the onset of scotoma, research has identified potential cortical reorganizations in both animal models (Shao et al., 2014; Burnat et al., 2017) and patients (Baker 2005), although the neural mechanisms driving these changes remain unclear. In this study, we developed deep neural networks with biological constraints, including foveated vision (see e.g., Jérémie et al., 2024), locally connected receptive fields and horizontal connections. These networks were pre-trained on images simulating the normal development of the visual system. Subsequently, the models were fine-tuned using masked images to investigate how the visual system adapts following scotoma onset. Post-training, the properties of these models were analyzed at the single-unit level using tools such as GradMaps (Rancon et al., 2025), and with population-level layer-wise approaches.



Simulating neurodegeneration in Deep Neural Networks with biological constraints

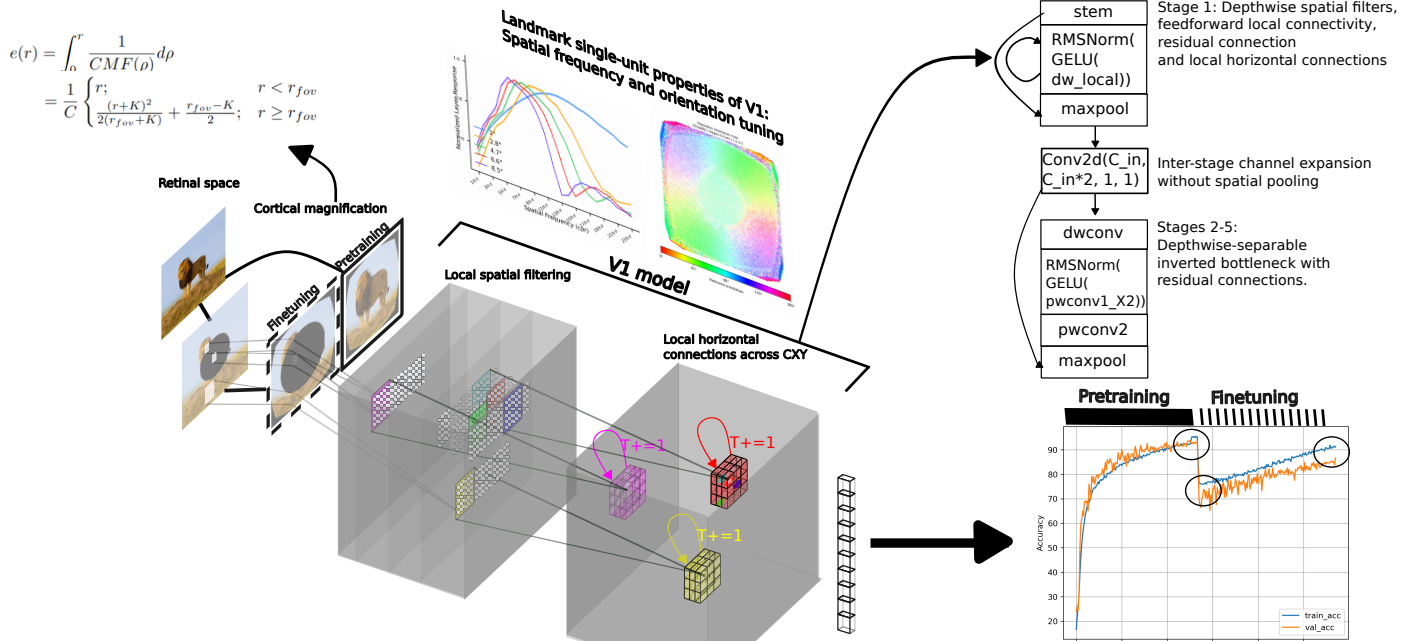
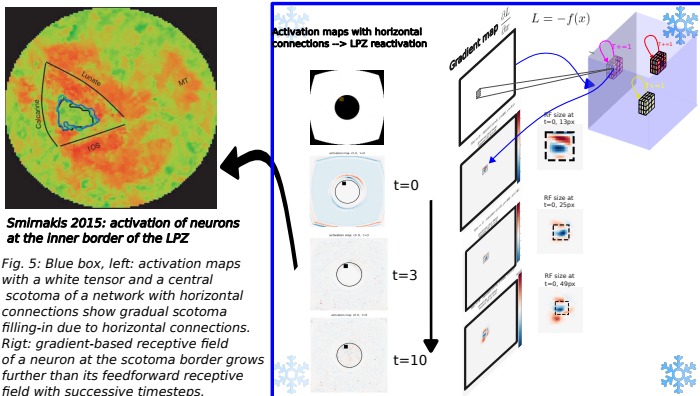


Fig. 1: Diagram of the training procedure and architecture that models the visual pathway's cortical magnification, local connectivity and horizontal connections. These result in landmark properties of V1: spatial frequency and orientation tuning, and allow for changes in receptive fields to be local and measurable.

Fig. 2: accuracy curves during healthy development and reorganization after an artificial scotoma

Measuring single-unit receptive field changes



Smirmakis 2015: activation of neurons at the inner border of the LPZ

Fig. 5: Blue box, left: activation maps with a white tensor and a central scotoma of a network with horizontal connections show gradual scotoma filling-in due to horizontal connections. Right: gradient-based receptive field of a neuron at the scotoma border grows further than its feedforward receptive field with successive timesteps.

Horizontal connections on the first layer of the network yield results that replicate the literature on filling-in after retinal lesions (McManus 2008, Gilbert 1995), single-unit receptive field growth and neuron reactivation at the LPZ border (Smirmakis 2015).

Measuring population-level and behavioural changes

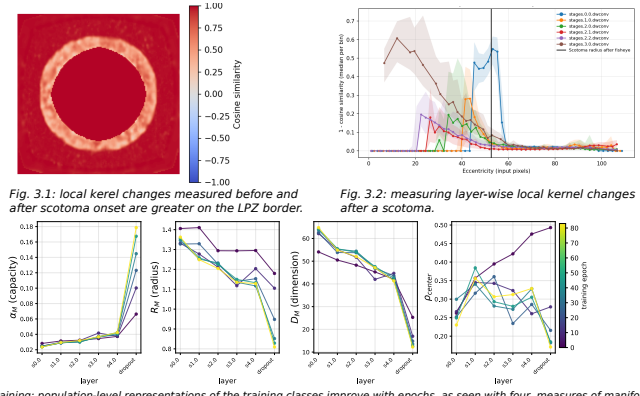


Fig. 3.1: local kernel changes measured before and after scotoma onset are greater on the LPZ border.

Fig. 3.2: measuring layer-wise local kernel changes after a scotoma.

Fig. 4: Pretraining: population-level representations of the training classes improve with epochs, as seen with four measures of manifold linear separability at each layer of the network. We aim to compare them with those at key stages of reorganization after scotoma onset.

Our locally connected architecture allows for measuring receptive field changes as a function of their eccentricity, which informs us of where in input space the local kernel changes (that in turn cause population representation changes) take place; these population representation measures can be correlated with accuracy improvements.

Next steps

1. Apply these methods to other visual pathologies, such as retinitis pigmentosa.
2. Tune the network's connectivity hyperparameters to those of the literature on healthy subjects and AMD patients.
3. Test other training procedures, including some based on self-supervised learning.
4. Use Spiking Neural Networks in order to increase the biological plausibility of the synaptic plasticity mechanism under study.
5. Train on event-based data.
6. Apply the system on computer vision applications for the reconstruction of occluded part of the visual scene.

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